$$A = \begin{pmatrix} 0 & -1 & 1 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix} \qquad P_A(x) \qquad \operatorname{grad} P_A = 4 \qquad P_A \in \mathbb{R}[x]$$

$$P_A(x) = \begin{vmatrix} -x & -1 & 1 & 0 \\ -1 & 1-x & 0 & 0 \\ 1 & 0 & -x & -1 \\ 0 & 0 & -1 & 1-x \end{vmatrix} = \begin{vmatrix} -x & \cdots \\ -x & \\ -x & \\ -x & \end{vmatrix} = (-x) \begin{vmatrix} 1 & -1 & 1 & 0 \\ 1 & 1-x & 0 & 0 \\ 1 & 0 & -x & -1 \\ 1 & 0 & -1 & 1-x \end{vmatrix} =$$

$$= (-x) \begin{vmatrix} 1 & -1 & 1 & 0 \\ 0 & 2-x & -1 & 0 \\ 0 & 1 & -x-1 & -1 \\ 0 & 1 & -2 & 1-x \end{vmatrix} = (-x) \begin{vmatrix} 2-x & -1 & 0 \\ 1 & -x-1 & -1 \\ 1 & -2 & 1-x \end{vmatrix} =$$

$$= (-x) \begin{vmatrix} 2-x & -1 & 0 \\ 1 & -x-1 & -1 \\ 0 & x-1 & 2-x \end{vmatrix} = (-x)\left((2-x)^2(-x-1) + (x-1)(2-x) + (2-x) \right) =$$

$$= (-x)(2-x)\left((x-2)(x+1) + (x-1) + 1\right)$$

$$= x(x-2)\left(x^2 - x - 2 + x - 1 + 1\right) = x(x-2)(x^2-2)$$

$0,\ 2,\ \sqrt{2},\ -\sqrt{2}$

$a_1 \dots a_n \in \mathbb{R}$ $\quad f : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ aplicația liniară a cărei matrice în baza canonică este $A = \begin{pmatrix} a_1 & a_2 & \dots & a_n \\ 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 1 \ \ 0 \end{pmatrix}$

a) $a_n \neq 0 \iff f$ izomorfism

b) $a_n = 0$, nu se știe care e baza pt $\ker(f)$ nici pt $\operatorname{Im}(f)$ și nu se verifică dacă $\ker(f) \oplus \operatorname{Im}(f) = \mathbb{R}^n$

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a) f izomorfism $\iff$ $\underbrace{M(f)}_{A}$ inversabilă $\iff \det A \neq 0$

$$\det A = \begin{vmatrix} a_1 & \dots & a_{n-1} & a_n \\ 1 & \ddots & & 0 \\ & \ddots & & \vdots \\ 0 & & 1 & 0 \end{vmatrix} = (-1)^{1+n} \cdot a_n \cdot \det I_{n-1} = (-1)^{n+1} a_n$$

b) $a_n = 0 \quad (\iff f$ nu e inversabilă$)$

$v \in \ker f \iff f(v) = 0$ (codomeniu)

$\mathcal{B}$, $\mathcal{B}'$ baze în $\mathbb{R}^n$, respectiv $\mathbb{R}^n$ (domeniu) $\Rightarrow [f(v)]_{\mathcal{B}'} = M(f)_{\mathcal{B}'}^{\mathcal{B}} [v]_{\mathcal{B}} \Rightarrow$

$\mathcal{B} = \mathcal{B}' = \mathcal{B}_c$ <u>bază canonică</u> din $\mathbb{R}^n$.

$v \in \mathbb{R}^n \quad v = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \Rightarrow [v]_{\mathcal{B}_c} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = v$

$= x_1 e_1 + \dots + x_n e_n$

$\Rightarrow f(v) = A \cdot v$

$f(v) = 0 \iff Av = 0 \Rightarrow$ $v = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$

$$\begin{cases} a_1 x_1 + \dots + a_{n-1} x_{n-1} = 0 \\ x_1 = 0 \\ x_2 = 0 \\ \dots \\ x_{n-1} = 0 \end{cases}$$

$\Rightarrow x_1 = \dots = x_{n-1} = 0, \quad x_n \in \mathbb{R}$

$$\Rightarrow \quad \ker(f) = \left\{ \begin{pmatrix} 0 \\ \vdots \\ 0 \\ x_n \end{pmatrix} \mid x_n \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix} \right\} \qquad \dim_{\mathbb{R}} \ker(f) = 1$$

$$\text{bază} \quad \left\{ \begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix} \right\}$$

$$\operatorname{Im}(f) = \{ w \in \mathbb{R}^n \mid \exists\, v \in \mathbb{R}^n .\ f(v) = w \}$$

$$w = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \qquad \exists\, v = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \qquad f(v) = w \iff A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$$\Rightarrow \begin{cases} a_1 x_1 + \dots + a_{n-1} x_{n-1} = y_1 \\ x_1 = y_2 \\ x_2 = y_3 \\ \dots \\ x_{n-1} = y_n \end{cases} \Rightarrow x_1 = y_2,\ x_2 = y_3,\ \dots,\ x_{n-1} = y_n,\ x_n \in \mathbb{R}$$

$$\Rightarrow a_1 y_2 + a_2 y_3 + \dots + a_{n-1} y_n = y_1$$

$$\Rightarrow \operatorname{Im}(f) = \left\{ \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n \mid y_1 = a_1 y_2 + \dots + a_{n-1} y_n \right\}$$

$$= Sp\left\{ \begin{pmatrix} a_1 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} a_2 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} a_{n-1} \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}$$

$n-1$ generatori

$$\dim_{\mathbb{R}} \mathrm{Im}(f) = n - \dim_{\mathbb{R}} \ker(f) = n-1 \Rightarrow \left\{ \begin{pmatrix} a_1 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} a_2 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} a_{n-1} \\ \vdots \\ 1 \end{pmatrix} \right\}$$

$a_1 e_1 + e_2$ $\quad$ $a_2 e_2 + e_3$ $\quad$ $a_{n-1} e_1 + e_n$

base di $\mathrm{Im}(f)$

$\ker(f) \oplus \mathrm{Im}(f) = \mathbb{R}^n$?

a) $\ker(f) \cap \mathrm{Im}(f) = \{0\}$?

b) $\ker(f) + \mathrm{Im}(f) = \mathbb{R}^n$?

a) Sia $v \in \ker(f) \cap \mathrm{Im}(f)$ $\quad$ $v \in \ker(f)$, $v \in \mathrm{Im}(f)$

$\Rightarrow \exists\ x_n \in \mathbb{R}$ a.t. $v = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ x_n \end{pmatrix}$

$\exists\ y_2 \dots y_n \in \mathbb{R}$ a.t. $v = \begin{pmatrix} a_1 y_2 + \dots + a_{n-1} y_n \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$

$$\Rightarrow \begin{cases} a_1 y_2 + \dots + a_{n-1} y_n = 0 \\ y_2 = 0 \\ y_3 = 0 \\ \dots \\ y_{n-1} = y_n = x_n \end{cases} \Rightarrow \begin{cases} y_2 = \dots = y_{n-1} = 0 \\ a_{n-1} y_n = 0 \\ y_n = x_n \end{cases}$$

$a_{n-1} = 0 \Rightarrow v = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ x_n \end{pmatrix} \Rightarrow \ker(f) \cap \operatorname{Im}(f) = \left\{ \begin{pmatrix} 0 \\ \vdots \\ 0 \\ x_n \end{pmatrix} \mid x_n \in \mathbb{R} \right\}$

$= \ker(f) \neq \{0\}$

$\Rightarrow$ soma $\ker(f) + \operatorname{Im}(f)$ não é direta

$\dim_{\mathbb{R}}(\ker(f) + \operatorname{Im}(f)) = \dim_{\mathbb{R}} \ker(f) + \dim_{\mathbb{R}} \operatorname{Im}(f) -$

$- \dim_{\mathbb{R}} \ker(f) \cap \operatorname{Im}(f) = n - 1$

$\Rightarrow \ker(f) + \operatorname{Im}(f) \neq \mathbb{R}^n$

$a_{n-1} \neq 0 \Rightarrow v = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \Rightarrow \ker(f) \cap \operatorname{Im}(f) = \{0\}$

$\dim_{\mathbb{R}} \ker(f) \cap \operatorname{Im}(f) = 0$

$$\Rightarrow \dim_{\mathbb{R}} \ker(f) \oplus \operatorname{im}(f) = \dim_{\mathbb{R}} \ker(f) + \dim_{\mathbb{R}} \operatorname{im}(f) = n \Rightarrow$$

$$\ker(f) \oplus \operatorname{im}(f) \subseteq \mathbb{R}^n$$

$$\Rightarrow \ker(f) \oplus \operatorname{im}(f) = \mathbb{R}^n$$